

Topic: Divergence & Curl

I. Divergence

Defn: Let $\mathbf{F}(x, y, z) = [M(x, y, z), N(x, y, z), P(x, y, z)]$ be a vector-valued function.
The **divergence** of $\mathbf{F}$, denoted $\text{div}(\mathbf{F})$, is given by

$$\text{div}(\mathbf{F}) =$$

Note: This can be remembered as: $\text{div}(\mathbf{F}) =$

where $\nabla =$ is the **del operator**.

Ex: Let $\mathbf{F}(x, y, z) = [0, \cos(xyz), \sin(xyz)]$. Find $\text{div}(\mathbf{F})$ at $P = (2, \frac{\pi}{2}, 0)$

Since

$$\Rightarrow \text{div}(\mathbf{F}) =$$

$$\Rightarrow \text{div}(\mathbf{F})(P) =$$

II. Curl

Defn: Let $\mathbf{F}(x, y, z) = [M(x, y, z), N(x, y, z), P(x, y, z)]$ be a vector-valued function.
The **curl** of $\mathbf{F}$, denoted $\text{curl}(\mathbf{F})$, is given by

$$\text{curl}(\mathbf{F}) =$$

Note: This can be remembered as: $\text{curl}(\mathbf{F}) =$

Ex: Let $\mathbf{F}(x, y, z) = [e^{-z^2}, e^{-x^2}, e^{-y^2}]$. Find $\text{curl}(\mathbf{F})$

III. Additional Concepts

Defn: Let $f(x, y, z)$ be a scalar-valued function. The **Laplacian** of f is

$$\operatorname{div}(\nabla f) =$$

Defn: Let $\mathbf{F}(x, y, z)$ be a vector-valued function. We say $\mathbf{F}$ is an **incompressible** (or **solenoidal**) vector field if

Defn: Let $\mathbf{F}(x, y, z)$ be a vector-valued function. We say $\mathbf{F}$ is an **irrotational** vector field if

Note: Irrotational field are

Thm: Let f be scalar-valued and $\mathbf{F}$ be vector-valued.

- Gradient fields are irrotational:
- The curl of a vector field is incompressible:

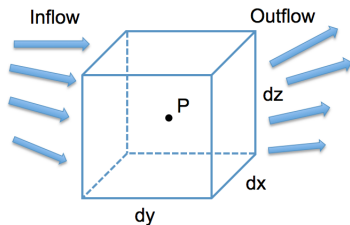
Ex: Let $\mathbf{v} = [0, 3z^2, 0]$ be a velocity field of some steady fluid flow. Answer the following:

- Is the flow incompressible?
- Is the flow irrotational?

Ex: Find the general form for $N(x, y, z)$ for which $\mathbf{F} = [-x \ln z, N, z^2]$ is incompressible?

IV. Physical Meaning

The Divergence at a point, P , can be thought of as a measurement of much flow goes in the box compared to how much flow goes out of the box (Net flow).



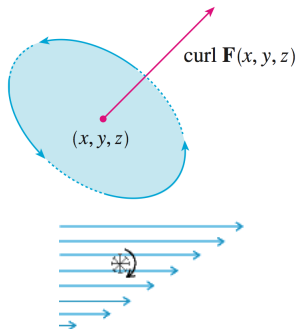
Divergence at $P \sim$ Net flow through a "small" box around P

If the outflow = inflow (net flow out = 0) then

If outflow $>$ inflow (net flow out $>$ 0) then

If outflow $<$ inflow (net flow out $<$ 0) then

The Curl at a point, P , measures the tendency for a flow to "curl" around a point.

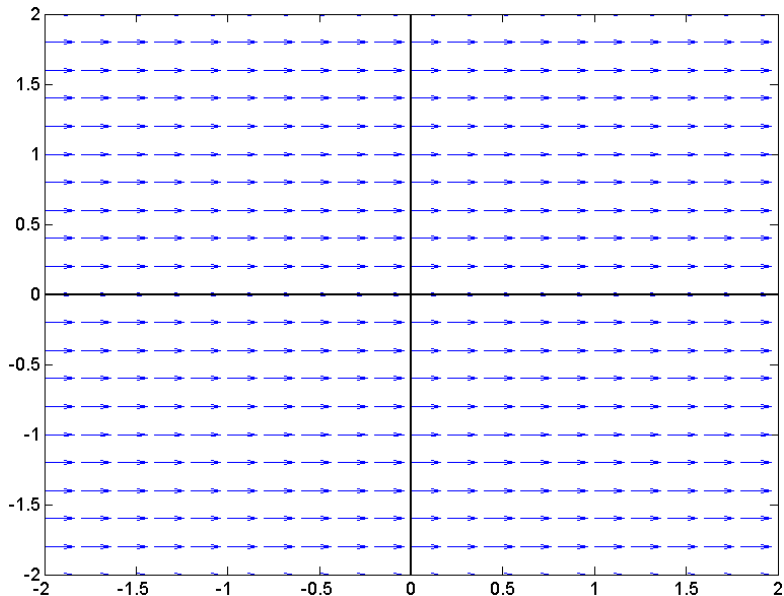


Flow acting counterclockwise around a point produces a curl pointing

Flow acting clockwise around a point produces a curl pointing

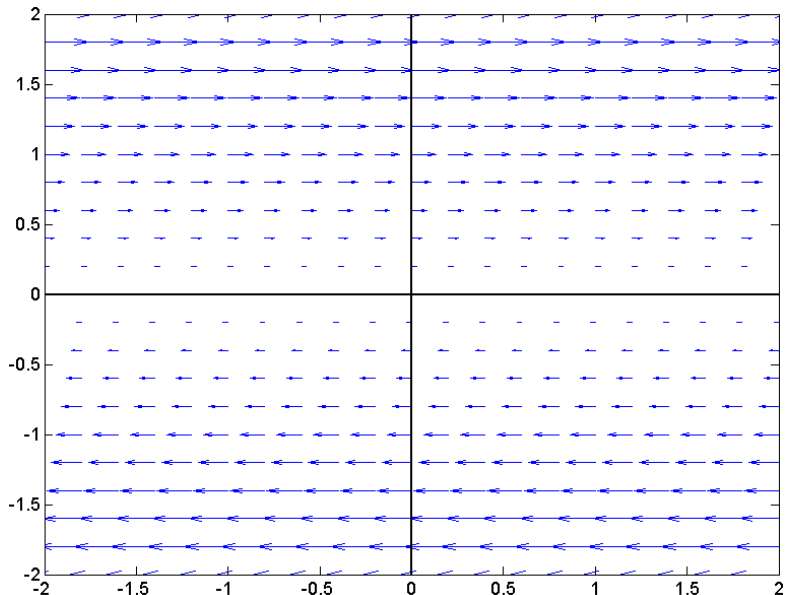
Ex: Let $\mathbf{F}$ be the flow field below. Is the flow...

- ...incompressible? If not, then is $\text{div}(\mathbf{F})$ positive or negative?
- ...irrotational? If not, then is $\text{curl}(\mathbf{F})$ pointing into or out of the board?



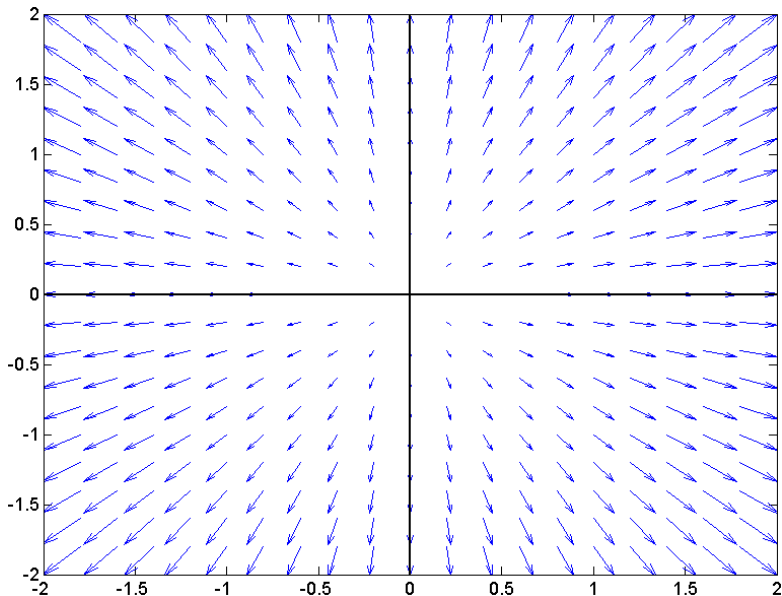
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V. Summary

- $\mathbf{F}$ conservative $\iff \mathbf{F} = \nabla f$ for some $f \iff \int_C \mathbf{F} \cdot d\mathbf{r}$ path independent $\iff \text{curl}(\mathbf{F}) = \mathbf{0}$
- $\mathbf{F}$ conservative & C closed $\implies \int_C \mathbf{F} \cdot d\mathbf{r} = 0$
- $\text{div}(\mathbf{F}) = \nabla \cdot \mathbf{F}$ (divergence)
- $\text{curl}(\mathbf{F}) = \nabla \times \mathbf{F}$ (curl)
- $\text{div}(\nabla f) = \nabla \cdot \nabla f = \nabla^2 f$ (Laplacian)
- $\text{div}(\mathbf{F}) = 0 \implies \mathbf{F}$ incompressible/solenoidal
- $\text{curl}(\mathbf{F}) = \mathbf{0} \implies \mathbf{F}$ irrotational
- $\text{curl}(\nabla f) = \mathbf{0} \longleftarrow \text{zero vector}$
- $\text{div}(\text{curl } \mathbf{F}) = 0$